RUSHIL GUPTA

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EDUCATION

Ashoka University, *B.Sc (Hons) in Mathematics and Computer Science with Research* Expected May 2027

- CGPA: 3.91/4, Major CGPA: 4.0/4.0.
- **Thesis:** Exploring loss of plasticity in continual learning, focusing on language models across training stages.
- **Relevant Coursework:** Reinforcement Learning, Machine Learning, Fourier Series, Linear Algebra and Matrix Analysis, Measure Theory, Advanced Algorithms, Cryptography, Stochastic Calculus and Mathematical Finance.
- **Teaching Assistant:** Reinforcement Learning, Introduction to ML, Numerical Algos & Optimization, Introduction to CS.

PUBLICATIONS

A Splitting Method for SDE Terminal-Law Estimation, *MLxOR Workshop, NeurIPS 2026* Under review at ICLR 2027
R. Gupta, S. Juneja

Fundamental Limits for Weighted Empirical Approximations of Exponentially Tilted Distributions, *AISTATS 2026*
S. R. Iyer, H. Mandal, D. Gupta, **R. Gupta**, A. Bandyopadhyay, A. Bassamboo, S. K. Juneja, V. Gupta

RESEARCH EXPERIENCE

Undergraduate Visiting Intern | Prof. Peter Glynn, Stanford University Jul 2026 – Aug 2026

- Investigating Empirical Bayes methods improving simulation estimators using Edgeworth expansions and deconvolution.

Undergraduate Researcher | Prof. Sandeep Juneja, Safexpress CDLDS Aug 2024 – Present

- Derived minimax allocations for SDE path-splitting estimators, reducing error by about 25% across several metrics.
- Studied sample-complexity thresholds for empirical exponential tilting, identifying distinct accuracy regimes.
- Explored diffusion models, training methods, and RL techniques as part of the center's diffusion research group.
- Developed pruning methods that use much smaller agent-based models to simulate disease spread in city-scale models.

Research Intern | Joint Cipher Bureau (DRDO) Jan 2026 – Present

- Studied the mathematical foundations of post-quantum cryptography, building up to identity-based encryption.
- Built an end-to-end PKI platform for offline post-quantum key distribution for hierarchical organizations.

Research Intern | KOITA Centre for Digital Health Ashoka Feb 2024 – Dec 2024

- Developed infrastructure for a cohort study integrating data streams from multiple wearables and a custom app.
- Worked on a solution to support informed-consent workflows, leveraging agentic LLMs for improved performance.
- Developed methods to identify high-quality subsets of ICU data, improving early mortality prediction accuracy by 10%.

WORK EXPERIENCE

ML Research Intern | Truth Audit Labs May 2026 – Jun 2026

- Generated ∼50K synthetic training samples for arbitrary natural-language policy classifiers, filtered by LLM judges.
- Fine-tuned a wide range of models across providers and scales (3B–30B) as classifiers; optimized inference with vLLM.
- Built an agentic red-teaming harness that composes prompt-injection attacks from past attempts and known jailbreaks.

Platform Team Intern | Sarvam AI Nov 2024 – Jan 2025

- Designed and implemented a versioning system for voice agents for incremental updates and seamless deployment.
- Part of the core team that owned the agents platform frontend and maintained several of its backend microservices.
- Wrote the first unit and integration tests for several backend microservices, improving API reliability.

POSITIONS OF RESPONSIBILITY

Head of Events | Ashoka Computer Science Society Sep 2024 – May 2025

- Started LeetCode Saturday, weekly competitive programming sessions for students to learn and solve problems together.
- Led a team of 20 members in planning and executing events, including talks, workshops, and mixers.

Co-Founder and Head of Events & Logistics | Ashoka Health Research Club Mar 2024 – Dec 2024

- Established a framework connecting students with faculty research, offering observership and internship tracks.

MISC

Reviewer, *ICLR 2027*

Squash Team Member, *Ashoka University* 2023 – Present